

## Numerical Assignment-02

1. Discuss the absolute and relative error. How to calculate the error percentile?

**Ans:**

**Absolute Error:** Absolute error quantifies the exact difference between a measured value and the true value. It provides a direct measure of how far off a measurement is, without considering the scale of the values involved.

Mathematical Expression:

$$\Delta x = |x - x_0|$$

where:

- $\Delta x$  is the absolute error.
- $x$  is the measured (or calculated) value.
- $x_0$  is the true value.

For example, if we measure a length as 15.5 cm when the actual length is 16.0 cm, the absolute error would be:

$$\Delta x = |15.5 \text{ cm} - 16.0 \text{ cm}| = 0.5 \text{ cm}$$

This means the measurement is off by 0.5 cm.

**Relative Error:** Relative error gives context to the absolute error by comparing it to the true value. This ratio helps us understand how significant the error is in relation to the actual measurement. It makes the error more meaningful, especially when comparing measurements that have different sizes or scales.

$$\text{Relative Error} = (|x - x_0| / x_0) * 100\%$$

We can use previous example/ $x_0$ \*100

## Error percentile

To calculate the error percentile:

1. **Collect Data:** Gather true values  $T$  and approximate values  $A$ .
2. **Calculate Relative Errors:**

$$\text{Relative Error}_i = \frac{|T_i - A_i|}{|T_i|}$$

3. **Sort Relative Errors:** Arrange errors in ascending order.
4. **Determine Percentile for  $e$ :**

$$\text{Percentile} = \left( \frac{\text{Count of errors} \leq e}{\text{Total errors}} \right) \times 100$$

### Example:

- True values: [10, 20, 30]
- Approximate values: [9, 22, 28]
- Relative errors: [0.1, 0.1, 0.0667] → Sorted: [0.0667, 0.1, 0.1]

For  $e = 0.1$ :

- Count: 2 errors  $\leq 0.1$
- Total: 3

$$\text{Percentile} = \left( \frac{2}{3} \right) \times 100 = 66.67\%$$

This indicates 66.67% of the errors are  $\leq 0.1$ .

This is the error percentile. It tells us how much your measurement deviates from the true value as a percentage of the true value itself.

2. For the equation  $y = 5\cos^3 x - e^{4x}$  find the root using the bisection method. Use the loop count up to 10 for breaking.

**Ans:**

To find the root of  $y = 5\cos^3 x - e^{4x}$  using the Bisection Method, follow these steps:

3. For the equation mentioned in 2, find the root using the Newton-Raphson method. Explain the factors of your initial guess.

When choosing an initial guess for the Newton-Raphson method, consider these factors:

1. **Graph Inspection:** Visualizing the function helps identify where it crosses the x-axis, indicating potential roots.
2. **Function Behavior:** Understand whether the function is increasing or decreasing around your guess to ensure you're moving toward a root.
3. **Previous Knowledge:** If you've solved similar equations before, use those known roots as a starting point.
4. **Sign Change:** Look for points where the function changes from positive to negative (or vice versa); this suggests a root is nearby.
5. **Close to the Root:** Select a guess close to where you think the root is located. This helps the method find the root faster.

4. What is a tabulated function? Describe a maxima and minima identification algorithm from a tabulated function.

**Ans:**

A **tabulated function** is a mathematical function that is represented in a discrete format, typically as a table of values. Each entry in the table consists of an input (or independent variable) and its corresponding output (or dependent variable). This representation is useful for functions that are difficult to express analytically or for empirical data collected from experiments. **Ex:**

| $x$ | $f(x)$ |
|-----|--------|
| 1   | 3      |
| 2   | 5      |
| 3   | 4      |
| 4   | 6      |
| 5   | 2      |

## Identifying Maxima and Minima from a Tabulated Function

To find the maxima (highest points) and minima (lowest points) from a tabulated function, you can use a simple algorithm that compares each function value with its neighboring values. This is often referred to as a neighbor comparison method.

### Algorithm Steps:

1. Input the table values:
  - Let the function be represented by pairs  $(x_1, f(x_1)), (x_2, f(x_2)), \dots, (x_n, f(x_n))$ , where  $x_1, x_2, \dots, x_n$  are the input values and  $f(x_1), f(x_2), \dots, f(x_n)$  are the corresponding output values.
2. Loop through the table:
  - Start from the second point and iterate through each point up to the second-last one, i.e., for  $i = 2$  to  $n - 1$ .
  - For each  $f(x_i)$ , compare it with its neighbors  $f(x_{i-1})$  and  $f(x_{i+1})$ .
3. Check for a local maximum or minimum:
  - Maxima: If  $f(x_i) > f(x_{i-1})$  and  $f(x_i) > f(x_{i+1})$ , then  $f(x_i)$  is a local maximum.
  - Minima: If  $f(x_i) < f(x_{i-1})$  and  $f(x_i) < f(x_{i+1})$ , then  $f(x_i)$  is a local minimum.
4. Handle boundary points:
  - For the first and last points  $(x_1, f(x_1))$  and  $(x_n, f(x_n))$ , check only one neighbor since they have fewer adjacent points.
    - First point: If  $f(x_1) > f(x_2)$ , it is a local maximum, and if  $f(x_1) < f(x_2)$ , it is a local minimum.
    - Last point: If  $f(x_n) > f(x_{n-1})$ , it is a local maximum, and if  $f(x_n) < f(x_{n-1})$ , it is a local minimum.
5. Output the maxima and minima:
  - List all the local maxima and minima points found in the table.

5. Elaborate the idea of local maxima and global maxima. Justify the statement “Newton's divided difference method always finds the global maxima”.

**Ans:**

### **Local Maxima and Global Maxima:**

- **Local Maxima:** A point  $x_i$  in the function  $f(x)$  is a local maximum if  $f(x_i) \geq f(x)$  for all  $x$  in some small neighbourhood around  $x_i$ . It is a peak within a specific region of the function.

**Example:** In  $f(x) = -x^2 + 4$ , at  $x=0$  (where  $f(0)=4$ ), it's higher than nearby points.

- **Global Maxima:** A point  $x^*$  is a global maximum if  $f(x^*) \geq f(x)$  for all  $x$  in the domain of the function. It is the highest point over the entire range of the function.

**Example:** Again, in  $f(x) = -x^2 + 4$ ,  $x=0$  is also the global maximum since it's the highest value for all  $x$ .

### **Justification for “Newton's divided difference method always finds the global maxima”:**

Newton's divided difference method **does not** always find the global maxima. In fact, this method is a numerical technique used for polynomial interpolation, not for finding maxima or minima of functions.

### **Key Points:**

1. **Purpose of the method:** Newton's divided difference method is used to find a polynomial that passes through a set of points. It helps in constructing an interpolating polynomial, which estimates unknown values from known data points.
2. **Not an optimization method:** Since the method is designed for interpolation, it does not inherently aim to find local or global maxima or minima of the function.
3. **Finding maxima:** To find the maxima of a function (especially a global maximum), you would typically use optimization methods such as:
  - Calculus (derivatives to find critical points)

- Numerical optimization algorithms (like Newton-Raphson, gradient descent, etc.)

Thus, Newton's divided difference method is not relevant to always finding the global maxima. It is simply a technique for constructing interpolating polynomials.

**6. Find the difference between convergence and divergence in the root finding algorithms. Are opening methods more prone to divergence? Explain.**

**Ans:**

In root-finding algorithms, **convergence** and **divergence** describe how the algorithm behaves as it tries to find the solution (root) of a function. Here's the difference between them:

**Convergence:**

- **Definition:** Convergence means that the algorithm gets closer and closer to the true root with each step.
- **What happens:** As the algorithm runs, the estimated value of the root improves over time, and eventually, it reaches or gets very close to the actual root.
- **Good result:** Convergence is what we want when using a root-finding algorithm because it means we are successfully finding the solution.

**Example:** If you use the Newton-Raphson method to find the root of a function and the guesses keep getting closer to the real root, the method is said to be converging.

**Divergence:**

- **Definition:** Divergence occurs when the algorithm moves away from the true root instead of approaching it.
- **What happens:** The estimated value drifts farther from the actual root or fails to stabilize, showing no progress toward a solution.
- **Bad result:** Divergence indicates the algorithm is not working, often due to poor initial guesses or unsuitability for the problem.

**Example:** In the Newton-Raphson method, if guesses move further from the real root or oscillate without converging, the method is diverging.

**Yes, some root-finding algorithms based on initial guesses, like open methods, can be more prone to divergence. Here's why:**

### **Open Methods:**

- **What are they:** Open methods, like the Newton-Raphson and Secant methods, start with one or two initial guesses and then iterate to find the root.
- **How they work:** These methods do not restrict the search to a specific interval (like closed methods). Instead, they use how the function behaves (like in Newton-Raphson) or earlier guesses (like in the Secant method) to help find the root.

### **Why they can diverge:**

1. **Depend on a good guess:** If the first guess is far from the root, the method can go in the wrong direction and fail to find the root.
2. **No boundaries:** Unlike closed methods (like the bisection method), open methods don't have limits on where they search, so they can easily move away from the root.
3. **Complex functions:** If the function has sharp curves or multiple roots, open methods might struggle and fail to get closer to the root.

7. Integrate  $f(x) = \sin^2 x - \cos^5 x + x^5 - x^4$  for the interval  $a=-5$  to  $b=5$  using Trapezoidal rule. Identify the error of the method.

## 8. Find the derivation of Simpson's 1/3rd rule of integration.

### Derivation of Simpson's 1/3 Rule

Simpson's 1/3 rule approximates the integral of a function using a quadratic polynomial. Here's the structured derivation:

---

#### 1. Dividing the Interval

Consider the interval  $[a, b]$  and divide it into  $n$  equal parts, where  $n$  is even:

$$h = \frac{b - a}{n}$$

The points are:

- $x_0 = a$
  - $x_1 = a + h$
  - $x_2 = a + 2h$
  - ...
  - $x_n = b$
- 

#### 2. Quadratic Polynomial Approximation

We want to approximate  $f(x)$  with a quadratic polynomial  $P(x)$  that passes through the points  $(x_0, f(x_0))$ ,  $(x_1, f(x_1))$ , and  $(x_2, f(x_2))$ .

Using Lagrange interpolation, the polynomial is expressed as:

$$P(x) = f(x_0) \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} + f(x_1) \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} + f(x_2) \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}$$

This polynomial is constructed to ensure that it matches the function values at  $x_0$ ,  $x_1$ , and  $x_2$ .

### 3. Integrating the Polynomial

Next, we integrate  $P(x)$  from  $a$  to  $b$ :

$$\int_a^b P(x) dx$$

Using the properties of definite integrals and the polynomial, we find:

$$\int_a^b f(x) dx \approx \frac{h}{3} \left( y_0 + y_n + 4 \sum_{\text{odd } i} f(x_i) + 2 \sum_{\text{even } i} f(x_i) \right)$$

where:

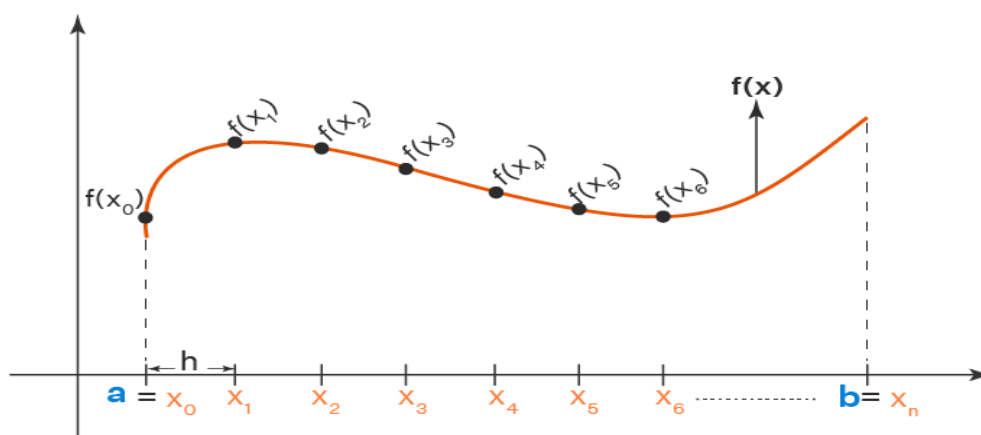
- $y_0 = f(x_0)$
  - $y_n = f(x_n)$
  - The odd-indexed terms contribute a factor of 4.
  - The even-indexed terms contribute a factor of 2.
- 

### 4. Final Formula

Thus, the formula for Simpson's 1/3 rule is:

$$\int_a^b f(x) dx \approx \frac{h}{3} \left( y_0 + y_n + 4 \sum_{\text{odd } i} f(x_i) + 2 \sum_{\text{even } i} f(x_i) \right)$$

The interval  $[a, b]$  is divided into subintervals of width  $h$



9. Discuss the comparative error of Simpson's  $\frac{1}{3}$  and  $\frac{3}{8}$  rules with example equation.

**Ans:**

### Simpson's $\frac{1}{3}$ Rule

- Formula:

$$\int_a^b f(x) dx \approx \frac{h}{3} (f(x_0) + 4f(x_1) + f(x_2))$$

where  $h = \frac{b-a}{2}$ , and  $x_0 = a$ ,  $x_1 = \frac{a+b}{2}$ , and  $x_2 = b$  divide the interval into two equal segments.

- Error Term:

$$E_t = -\frac{(b-a)^5}{2880} f^{(4)}(\xi)$$

for some  $\xi \in (a, b)$ . The error term indicates that the accuracy of the rule depends on the fourth derivative of the function  $f(x)$  and the interval width  $(b-a)$ .

---

### Simpson's $\frac{3}{8}$ Rule

- Formula:

$$\int_a^b f(x) dx \approx \frac{3h}{8} (f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3))$$

where  $h = \frac{b-a}{3}$ , and the points  $x_0, x_1, x_2, x_3$  divide the interval into three equal parts.

- Error Term:

$$E_t = -\frac{(b-a)^5}{6480} f^{(4)}(\xi)$$

for some  $\xi \in (a, b)$ . This error term shows that the Simpson's  $\frac{3}{8}$  rule is more accurate than the Simpson's  $\frac{1}{3}$  rule due to the smaller constant in the denominator.

## Example

Let's take a simple function to compare the two methods:

$$f(x) = x^4$$

over the interval  $[0, 1]$ .

### Calculating Integrals:

#### 1. Exact Integral:

$$\int_0^1 x^4 dx = \left[ \frac{x^5}{5} \right]_0^1 = \frac{1}{5} = 0.2$$

#### 2. Using Simpson's 1/3 Rule:

- Let  $n = 2$  (two intervals):
- $h = \frac{1-0}{2} = 0.5$
- $x_0 = 0, x_1 = 0.5, x_2 = 1$
- Function values:
  - $f(x_0) = 0^4 = 0$
  - $f(x_1) = (0.5)^4 = 0.0625$
  - $f(x_2) = 1^4 = 1$

Applying Simpson's 1/3:

$$T_{1/3} = \frac{0.5}{3} (0 + 4(0.0625) + 1) = \frac{0.5}{3} (0 + 0.25 + 1) = \frac{0.5 \cdot 1.25}{3} = \frac{0.625}{3} \approx 0.2083$$

### 3. Using Simpson's 3/8 Rule:

- For  $n = 3$  (three intervals):
- $h = \frac{1-0}{3} \approx 0.3333$
- $x_0 = 0, x_1 \approx 0.3333, x_2 \approx 0.6667, x_3 = 1$
- Function values:
  - $f(x_0) = 0$
  - $f(x_1) \approx (0.3333)^4 \approx 0.0123$
  - $f(x_2) \approx (0.6667)^4 \approx 0.1975$
  - $f(x_3) = 1$

Applying Simpson's 3/8:

$$T_{3/8} = \frac{0.3333}{3} (0 + 3(0.0123) + 3(0.1975) + 1) \approx \frac{0.3333}{3} (0 + 0.0369 + 0.5925 + 1) \approx \frac{0.3333}{3} (0.6294) \approx 0.2098$$

### Summary of Errors:

- Exact Integral: 0.2
- Simpson's 1/3 Result: 0.2083 (Error: 0.0083)
- Simpson's 3/8 Result: 0.2098 (Error: 0.0098)

In this case, the 1/3 Rule gives a slightly better approximation than the 3/8 Rule for this specific function, but the error behavior and constants show the 3/8 Rule generally can be advantageous for more complex functions or higher polynomial degrees.

10. Given the integral  $\int_0^6 (x^5 - 3x^4 + 5x) dx$ , use Simpson's 3/8 rule to approximate the value of integral. Divide the  $[0, 6]$  into three equal subintervals. Show all steps in your calculation and compare the result with the exact value of the integral.

**Ans:**

To approximate the integral

$$\int_0^6 (x^5 - 3x^4 + 5x) dx$$

using Simpson's 3/8 rule, we will divide the interval  $[0, 6]$  into three equal subintervals. Here are the steps:

### Step 1: Define the Function

Let

$$f(x) = x^5 - 3x^4 + 5x$$

### Step 2: Determine the Subintervals

The interval  $[0, 6]$  is divided into three equal parts:

- $a = 0$
- $b = 6$
- Number of subintervals  $n = 3$  (which is a multiple of 3 for the 3/8 rule)
- Width of each subinterval:

$$h = \frac{b - a}{n} = \frac{6 - 0}{3} = 2$$

### Step 3: Calculate the Points

The points are:

- $x_0 = 0$
- $x_1 = 2$
- $x_2 = 4$
- $x_3 = 6$

#### Step 4: Evaluate the Function at Each Point

Now, calculate  $f(x)$  at each point:

- $f(0) = 0^5 - 3(0)^4 + 5(0) = 0$
- $f(2) = 2^5 - 3(2^4) + 5(2) = 32 - 48 + 10 = -6$
- $f(4) = 4^5 - 3(4^4) + 5(4) = 1024 - 768 + 20 = 276$
- $f(6) = 6^5 - 3(6^4) + 5(6) = 7776 - 3888 + 30 = 3918$

#### Step 5: Apply Simpson's 3/8 Rule

The formula for Simpson's 3/8 rule is:

$$\int_a^b f(x) dx \approx \frac{3h}{8} [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)]$$

Substituting our values:

$$\int_0^6 f(x) dx \approx \frac{3 \times 2}{8} [f(0) + 3f(2) + 3f(4) + f(6)]$$

Calculating the terms:

$$= \frac{6}{8} [0 + 3(-6) + 3(276) + 3918]$$

Calculating inside the brackets:

$$= \frac{6}{8} [0 - 18 + 828 + 3918]$$

$$= \frac{6}{8} [4828]$$

$$= \frac{6 \times 4828}{8} = \frac{28968}{8} = 3621$$

### Step 6: Exact Value of the Integral

To find the exact value of the integral, we compute:

$$\int_0^6 (x^5 - 3x^4 + 5x) dx$$

Calculating the antiderivative:

$$F(x) = \frac{x^6}{6} - \frac{3x^5}{5} + \frac{5x^2}{2}$$

Now evaluate from 0 to 6:

$$F(6) = \frac{6^6}{6} - \frac{3(6^5)}{5} + \frac{5(6^2)}{2}$$

$$= 7776 - \frac{3 \times 7776}{5} + 90$$

$$= 7776 - 4665.6 + 90 = 3100.4$$

$$F(0) = 0$$

Thus, the exact value is:

$$\int_0^6 (x^5 - 3x^4 + 5x) dx = 3100.4$$

### Step 7: Comparison

- **Approximate Value (Simpson's 3/8):** 3621
- **Exact Value:** 3100.4

### Conclusion

The Simpson's 3/8 rule approximation is 3621, while the exact integral value is 3100.4. The approximation is close but slightly higher than the exact value.